

FILE : CH11PP1

```
lst = [0,1]
a = 0
b = 1
c = 0
for i in range(7):
    c = a + b
    a = b
    b = c
    lst.append(c)
print("9 terms of Fibonacci series are:", tuple(lst))
```

#2

```
print()
print("#2")
tup = ()
a = 0
b = 1
c = 0
tup=tup+(a,)
tup=tup+(b,)
for i in range(7):
    c = a + b
    a = b
    b = c
    tup=tup+(c,)
print("9 terms of Fibonacci series are:", tup)
```

FILE : CH11PP1A

Fibonacci series using 2 variable only

```
a = 0
b = 1
print(a,b,end=" ")
for i in range(7):
    b = a + b
    print(b,end=" ")
    a=b-a
```

```
#2
print()
print("#2")
a = 0
b = 1
lst=[a,b]
for i in range(7):
    b = a + b
    lst.append(b)
    a=b-a
print("9 terms of Fibonacci series are:", lst)
```

#3

```
print()
print("#3")
a = 0
b = 1
lst=[a,b]
for i in range(7):
    b = a + b
    lst.append(b)
    a=b-a
print("9 terms of Fibonacci series are:", tuple(lst))
```

FILE : CH11PP2

```
tup = eval(input("Enter the elements of tuple:"))
b = int(eval(input("Enter the index value:")))
c = len(tup)
if b < c:
    print("value of tuple at index", b ,"is:" ,tup[b])
else:
    print("Index is out of range")
```

#2b

```
term = int(input ("Enter Fibonacci Term: "))
lst = [0,1]
a = 0
b = 1
c = 0
for i in range(term+1):
    c = a + b
    a = b
    b = c
    lst.append(c)
tup=tuple(lst)
print("Terms of Fibonacci series are:",tup)
ele=eval(input("Enter the index value:"))
print("index of ",ele,"is ",tup.index(ele))
```

FILE : CH11PP2A

```
lst = [0,1]
a = 0;b = 1;c = 0
for i in range(7):
    c = a + b
    a = b;b = c
    lst.append(c)
tup=tuple(lst)
print("Terms of Fibonacci series are:",tup)
ele=eval(input("Enter the index value:"))
print("index of ",ele,"is ",tup.index(ele))
```

FILE : CH11PP3

```
n = eval(input("Enter the numbers: "))
tup = tuple(n)
print("Tuple is:", tup)
print("Highest value in the tuple is:", max(tup))
print("Lowest value in the tuple is:", min(tup))
```

FILE : CH11PP4

```
tup = ()
ans = "y"
while ans == "y" or ans == "Y" :
    roll_num = int(input("Enter roll number of student: "))
    name = input("Enter name of student: ")
    marks = int(input("Enter marks of student: "))
    tup += ((roll_num, name, marks),)
    ans = input("Do you want to enter more marks? (y/n): ")
print(tup)
```

FILE : CH11PP5

```
num_of_students = 5
tup = ()
for i in range(num_of_students):
    print("Enter the marks of student", i + 1)
    m1 = int(input("Enter marks in first subject: "))
    m2 = int(input("Enter marks in second subject: "))
    m3 = int(input("Enter marks in third subject: "))
    tup = tup + ((m1, m2, m3),)
    print()
print("Nested tuple of student data is:", tup)
```

FILE : CH11PP6

```
num_of_students = 5
tup = ()
for i in range(num_of_students):
    print("Enter the marks of student", i + 1)
    m1 = int(input("Enter marks in first subject: "))
    m2 = int(input("Enter marks in second subject: "))
    m3 = int(input("Enter marks in third subject: "))
    s=(m1,m2,m3)
    tup = tup + (s,)
    s1=sum(s)
    print("total of student ",i," is ",s1)
    print("average of student ",i," is ",s1/len(s))
    print()
print("Nested tuple of student data is:", tup)
```

FILE : CH11PP6A

```
num_of_students = 5
tup = ()
tup1=()
for i in range(num_of_students):
    print("Enter the marks of student", i + 1)
    m1 = int(input("Enter marks in first subject: "))
    m2 = int(input("Enter marks in second subject: "))
    m3 = int(input("Enter marks in third subject: "))
    s=(m1,m2,m3)
    tup = tup + (s,)
    s1=sum(s)
    tup1+=(s1,)
print()
print("Nested tuple of student data is:", tup)
print()
for i in range(num_of_students):
    print("total of student ",i," is ",tup1[i])
    print("average of student ",i," is ",tup1[i]/len(s))
    print()
```

FILE : CH11PP6B

```
num_of_students = 5
tup = ()
tup1=()
for i in range(num_of_students):
    print("Enter the marks of student", i + 1)
    m1 = int(input("Enter marks in first subject: "))
    m2 = int(input("Enter marks in second subject: "))
    m3 = int(input("Enter marks in third subject: "))
    s=(m1,m2,m3)
    tup = tup + (s,)
    s1=sum(s)
    tup1+=(s1,)
print()
print("Nested tuple of student data is:", tup)
print()
for i in range(num_of_students):
    print("elements of tuple ",i+1," ",tup[i])
    print()
print()
for i in range(num_of_students):
    print("total of student ",i," is ",tup1[i])
    print("average of student ",i," is ",tup1[i]/len(s))
    print()
```

FILE : CH11PP7

```
tup1 = eval(input("Enter the elements of first tuple: "))
tup2 = eval(input("Enter the elements of second tuple: "))
tup3 = tup1 + tup2
print(tup3)
```

FILE : CH11PP8

```
tup = ()
for i in range(1,51):
    tup = tup + (i**2,)
print("The square of integers from 1 to 50 is:" ,tup)
```

#b

```
print()
tup = ()
for i in range(1, 27):
    tup = tup + (chr(i + 96)* i,)
print(tup)
```

FILE : CH11PP9

```
tup = ()
for i in range(1,51):
    tup = tup + (i**2,)
print("The square of integers from 1 to 50 is:" ,tup)
```

#b

```
print()
tup = ()
for i in range(1, 27):
    tup = tup + (chr(i + 96)* i,)
print(tup)
```

FILE : CH11PP10

```
tup = ((2,5),(4,2),(9,8),(12,10))
count = 0
tup_length = len(tup)
for i in range (tup_length):
    if tup [i][0] % 2 == 0 and tup[i][1] % 2 == 0:
        count = count + 1
print("The number of pair where both a and b are even:", count)
```

FILE : CH11PP11

```
seq_a = eval(input("Enter the first tuple: "))
seq_b = eval(input("Enter the second tuple: "))
for i in seq_a:
    if i not in seq_b:
        print("False")
        break
else:
    print("True")
```

FILE : CH11PP12

```
tup = eval(input("Enter the numeric tuple: "))
total = sum(tup)
tup_length = len(tup)
mean = total / tup_length
print("Mean of tuple:", mean)
```

FILE : CH11PP13

```
tup = eval(input("Enter a tuple: "))
maxCount = 0
mode = 0
for i in tup :
    count = tup.count(i)
    if maxCount < count:
        maxCount = count
        mode = i

print("mode:", mode)
```

FILE : CH11PP14

```
import statistics
tup = eval(input("Enter a tuple: "))
tup_sum = sum(tup)
tup_len = len(tup)
print("Average of tuple element is:", tup_sum / tup_len)
print("Mean of tuple element is:", statistics.mean(tup))
```

FILE : CH11PP15

```
tup1 = ((1, 2), (3, 4.15, 5.15), ( 7, 8, 12, 15))
total_mean = 0
tup1_len = len(tup1)
for i in range(tup1_len):
    mean = sum(tup1[i]) / len(tup1[i])
    print("Mean element", i + 1, ":", mean)
    total_mean = total_mean + mean
print("Mean of means" ,total_mean / tup1_len)
```